

An explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of user reviews.

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1. Title

Explainable Feature-Based Smartphone Recommendation System Using Aspect-Based Sentiment Analysis of User Reviews

2. Background of Research

2.1 Introduction to Recommendation Systems

Today, Selecting a smartphone is more harder than it looks. With thousands of models competing across a dozen price brackets, most buyers spend hours scrolling through review pages, watching at star ratings, and still keep uncertain. The problem is not a lack of information , it is that current recommendation systems are not built to make sense of it. Traditional approaches tend to surface phones that other users bought or that carry high average ratings, without ever examining what those buyers actually said about the device. Whether 80% of reviewers loved the camera and quietly tolerated a weak battery, or the reverse, gets collapsed into a single number that tells us almost nothing useful.

This research proposes a different way of thinking about the problem. Rather than treating overall ratings as the primary signal, the system works through thousands of real user reviews and analyses them feature by feature. It can, for instance, determine that 87% of reviewers wrote positively about a particular phone's camera while only 54% were satisfied with its battery life. Those granular, feature-level sentiment scores become the building blocks for matching phones to what each individual user genuinely values.

The mechanism behind this is a feature-based recommendation scoring formula. Each aspect like camera, battery, display, performance, price receives a sentiment score derived from how buyers talk about it in their reviews. When a user specifies their own priorities (say, 40% weight on battery and 30% on camera), the system multiplies each feature score by its assigned weight and sums the results into a final rank for every phone. A device with strong battery feedback might score $0.85 \times$

0.4 = 0.34 for that dimension alone, before the other features are factored in, producing an overall rank that reflects the user's actual needs rather than general popularity.

This matters most in a market where no single device excels at everything. A photography enthusiast and a mobile gamer are looking for fundamentally different things, and a recommendation system worth using should recognize that. By bringing together feature-level sentiment analysis and explicit user preference weighting, this research sets out to build a system that offers clear, honest, personalized recommendations and explains in plain language why each phone was suggested.

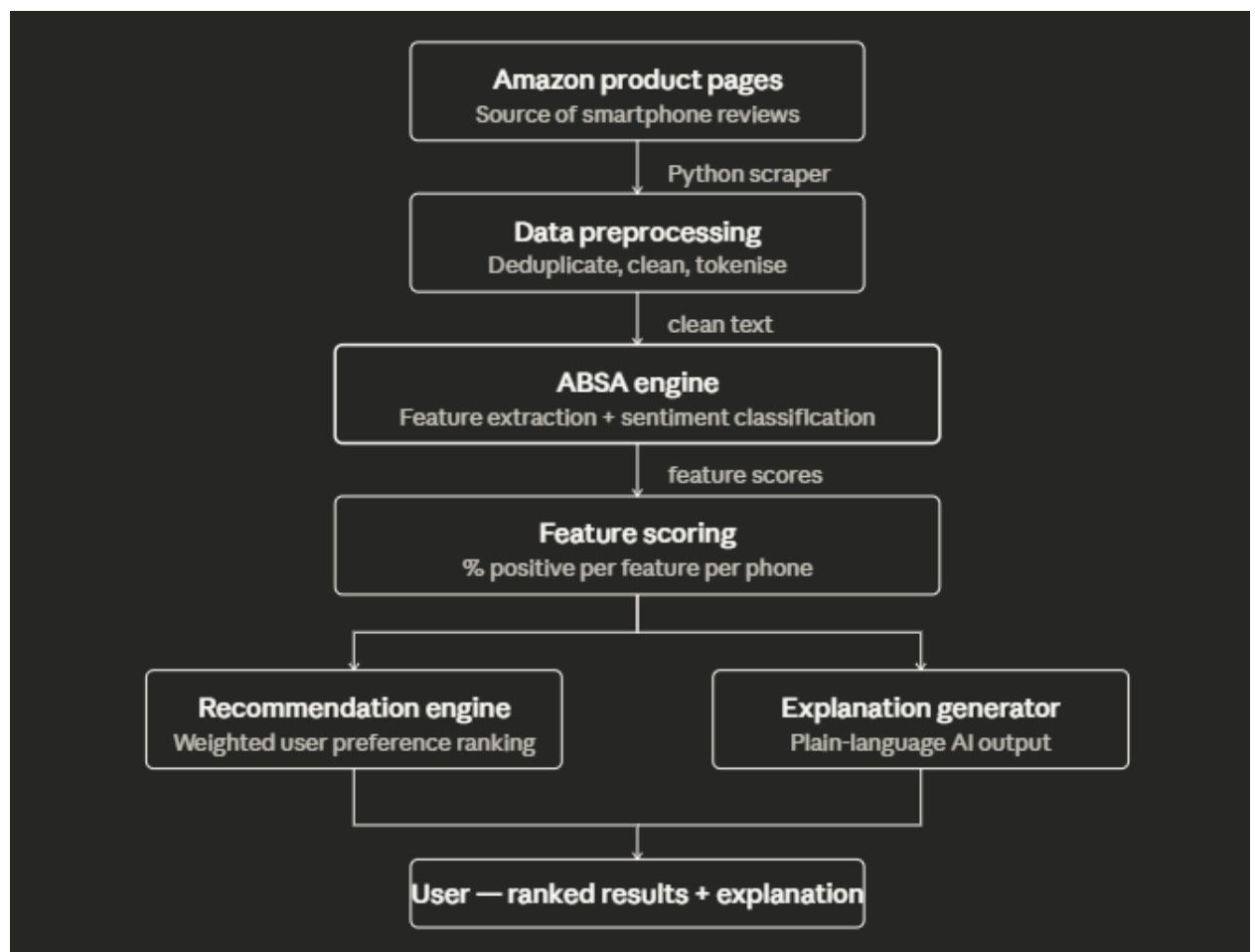


Figure 1-Overall System Architecture

2.2 Limitations of Traditional Recommendation Systems

For all their popularity, traditional recommender systems carry a set of well-documented weaknesses that limit how useful they can actually be. The most fundamental is their dependence on interaction history. **Collaborative filtering**- one of the most widely used approaches works by identifying patterns in past ratings, clicks, and purchases across users. When that historical data is thin or missing entirely, the system has little to go on and performance drops noticeably.

This is what gives rise to the cold-start problem. A new user signing up for a platform, or a newly launched product with no review history, cannot be handled well by these systems because there is simply no prior data from which to draw inferences. Recommendations in these cases tend to be generic at best and irrelevant at worst.

The absence of explainability is another serious limitation. Most traditional recommendation models work as **black boxes** they output a ranked list of products without offering any window into the reasoning that produced it. Users have no way of knowing whether a recommendation reflects something they actually care about, which understandably erodes trust in the system over time.

Perhaps most relevant to this research is the reliance on aggregated ratings. When a system reduces an entire product experience to a single average score, it necessarily discards the detail that matters most to informed decision-making. A smartphone may hold a strong overall rating despite widespread complaints about its battery or camera performance. Those specific opinions get averaged away and never surface in the recommendation. This is precisely the limitation that Aspect-Based Sentiment Analysis is designed to address, and the gap that motivates the approach taken here.

2.3 Role of User Reviews in Product Evaluation

The explosion of e-commerce platforms like Amazon and eBay has quietly but fundamentally shifted how people decide what to buy. Where shoppers once relied on manufacturer descriptions and spec sheets, they now turn first to what other buyers have to say. Peer opinion has become the most trusted currency in online purchasing and for good reason. A stranger who owns the product and has lived with it for three months carries far more credibility than any promotional copy written

to sell it. Reviews written by actual buyers carry the weight of real-world experience, and research consistently shows that they exert a strong influence on purchasing confidence. In this sense, online reviews function as a modern, digital form of word-of-mouth (e-WOM), and their importance to the consumer decision process is well established.

What makes review analysis genuinely valuable is its capacity to convert unstructured opinion into structured, actionable knowledge. A four-star rating tells a potential buyer that most people were broadly The review text itself tells a much richer story. Buyers write about how long the battery actually lasts through a heavy day, how the camera handles a dimly lit restaurant, whether the phone feels solid in the hand or plasticky, how snappy the interface feels after six months of use. None of that nuance survives compression into a single star rating.

Reviews also have a way of exposing what high ratings conveniently hide. A phone can sit comfortably at 4.3 stars while quietly accumulating complaints about throttling under sustained load or a camera that falls apart the moment the sun goes down. Users who stop at the aggregate score never see any of that and that gap is exactly what this research sets out to close.

Users relying on the star rating alone would never know. Aspect-Based Sentiment Analysis is designed precisely to expose these hidden dynamics by identifying each specific feature mentioned in a review and assessing the sentiment directed at it individually, ABSA produces a nuanced, accurate picture of a product's real strengths and weaknesses.

For this study, smartphone reviews collected from Amazon form the primary dataset. Amazon was selected because it offers one of the largest and most diverse collections of genuine, English-language consumer electronics reviews available anywhere, spanning all major manufacturers and price points. Systematically scraping, cleaning, and analyzing this data allows the system to extract feature-level sentiments and consolidate them into structured performance scores. What emerges is a recommendation tool built entirely on what real people experienced not what a brand wanted buyers to believe, and not which product happened to trend that week.

2.4 Introduction to Aspect-Based Sentiment Analysis (ABSA)

Aspect-Based Sentiment Analysis goes well beyond what conventional sentiment classification can do within natural language processing. Where standard sentiment analysis assigns a single positive, negative, or neutral label to an entire piece of text, ABSA breaks that text apart and evaluates how the writer feels about each individual feature or aspect of the product being discussed.

Take a review that says: "The display is stunning, but I am genuinely disappointed with the battery life." A standard sentiment classifier looks at that sentence as a whole and likely lands on neutral the positive and negative cancel each other out and the actual meaning gets lost in the averaging. ABSA correctly identifies that the display is receiving enthusiastic praise while the battery is drawing real dissatisfaction. That distinction is exactly what a feature-driven recommendation system needs in order to be useful.

In this research, ABSA is applied to six smartphone features that consistently appear in consumer reviews and that buyers most commonly weigh when choosing a device: camera, battery, display, performance, design, and price. Each of these aspects is treated as an independent dimension of evaluation, with sentiment classified separately for every mention across the entire review dataset.

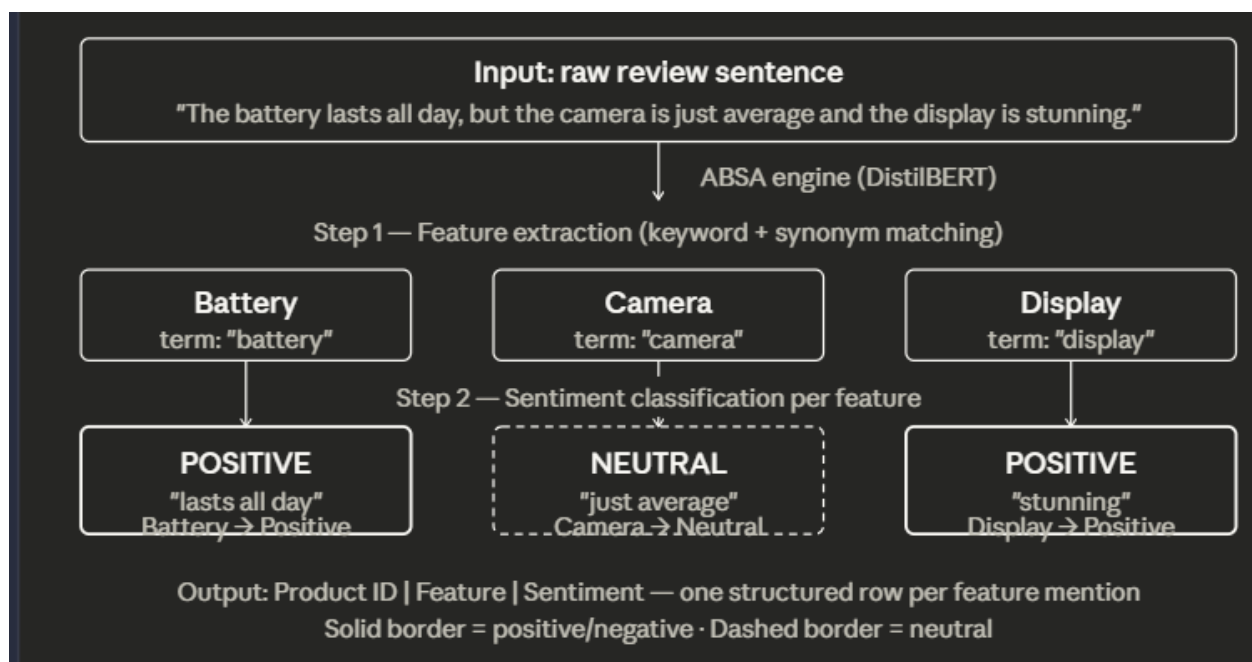


Figure 2-ABSA process: from review to feature sentiment

2.5 Explainable AI and Why It Matters

Building a system that analyses user reviews at the feature level is a technically ambitious goal, and the path to getting there involves several real challenges. The first is the nature of review text itself. People write in natural language – informal, idiosyncratic, and inconsistent. Spelling mistakes, abbreviations, slang, and sentences that bundle opinions about two different aspects together are all common. Teaching a system to reliably pick out predefined features such as camera or battery from this kind of **unstructured text demands robust NLP techniques** that can handle variability without losing accuracy.

Managing the quality of the underlying data adds another layer of complexity. Large e-commerce review datasets are rarely clean. One common issue is reviewers posting the exact same text across multiple product pages, which artificially inflates sentiment counts and pulls scores away from reality. Spam entries, copy-pasted comments, and near-identical submissions pile on top of that, muddying the dataset further. Catching and removing these duplicates by cross-checking both the product identifier and the review content is essential but it has to be done carefully, because an overly aggressive filter will end up stripping out legitimate reviews along with the junk.

Getting the sentiment classification right at the feature level is arguably the hardest part. Unlike document-level sentiment analysis, feature-level classification must correctly associate each sentiment expression with the right aspect within a single sentence. A review that says "The display is great, but the battery is disappointing" requires the model to assign a positive label to display and a negative label to battery independently. Achieving this with high reliability requires either well-designed linguistic rules or a sufficiently powerful machine learning model – ideally both working in combination.

Finally, converting feature-level sentiment classifications into meaningful, comparable performance scores is more complex than it first appears. The raw counts of positive and negative mentions must be normalized in a way that remains fair across phones with very different numbers of reviews. A device with five hundred reviews and a 90% positive camera sentiment is not directly comparable to one with fifty reviews and the same percentage, without accounting for statistical confidence. Designing a scoring mechanism that is both intuitive and statistically sound is a central methodological concern of this research.

2.6 Problem Domain Significance

Few consumer markets move as quickly or generate as much complexity as the smartphone industry. Apple, Samsung, Xiaomi, and a host of other manufacturers release new models on overlapping cycles, each one differentiated by increasingly subtle variations in specification, design language, and targeted use case. For the average consumer trying to make a purchasing decision, this creates a genuine problem: too many options, too much information, and no easy way to filter either based on what actually matters to them personally.

The volume of review content compounds the challenge. A popular smartphone model on Amazon might accumulate thousands of individual reviews, each one containing observations about different aspects of the device's performance. Reading through a meaningful fraction of that content manually is not a realistic option for most buyers, so purchasing decisions end up being driven by overall star ratings, brand familiarity, or the most recent marketing campaign none of which reliably reflect whether a given device fits a specific user's requirements.

What is increasingly clear is that users want more than a ranked list. Especially for high-involvement purchases like smartphones, people want to understand why a particular device is being recommended to them. This expectation is pushing the field toward Explainable Artificial Intelligence systems that make their reasoning visible and interpretable rather than hiding it behind a prediction score.

This research is positioned squarely at that intersection. By combining feature-level sentiment analysis with a transparent, user-driven recommendation framework, it offers a way to turn the enormous volume of available review content into something genuinely decision-relevant giving buyers real insight into what other users have actually experienced, matched directly to the features they care about most.

3. Literature Review

This section reviews existing literature related to recommendation systems, sentiment analysis, and explainable AI, with particular attention to Aspect-Based Sentiment Analysis (ABSA) and its application in product recommendation contexts. The review is organized thematically covering foundational recommendation approaches, sentiment analysis techniques, review-driven systems, and explainable AI before identifying the specific research gaps this study aims to address. Together, these areas establish the theoretical and empirical foundation for the system proposed here.

3.1 Recommendation Systems

Recommendation systems have become core infrastructure for e-commerce platforms, helping users navigate large product catalogues by surfacing items that match their apparent preferences and past behaviors. The field has historically been organized around three main approaches: **collaborative filtering** (CF), which draws on patterns in user item interaction data such as ratings and purchase history; **content-based filtering** (CBF), which matches item attributes directly to user preference profiles; and hybrid models that blend both strategies to gain the advantages of each.

Each of these approaches carries well-known limitations. Collaborative filtering is especially vulnerable to data sparsity and the cold-start problem when a new user joins the platform or a new product is added, there is simply not enough interaction data to generate reliable recommendations. Content-based filtering, for its part, tends toward over-specialization, repeatedly surfacing items similar to what a user has already engaged with and failing to introduce genuinely novel suggestions.

More recent work has focused on integrating deep learning techniques to push past these constraints. A hybrid recommendation model developed by Sami et al. (2024), for instance, layers together user-based and item-based collaborative filtering, Neural Collaborative Filtering (NCF), Recurrent Neural Networks (RNN), and TF-IDF-based content analysis to improve recommendation quality substantially. This kind of integrated architecture allows the system to

capture both linear and non-linear relationships in user–item interactions as well as temporal patterns in behavior, with strong results across standard evaluation metrics including RMSE, MAE, Precision, and Recall.

Even so, deep learning-based hybrid systems bring their own difficulties. They typically demand large-scale training data, significant computational resources, and careful hyperparameter tuning. More importantly for the purposes of this research, they tend to operate as black-box models producing recommendations without any transparent reasoning the user can inspect or challenge. That opacity is precisely the problem this study sets out to address: the need for recommendation systems that are not only accurate but also interpretable and genuinely informative.

3.2 Sentiment Analysis and ABSA

Sentiment analysis is a long-established branch of natural language processing concerned with identifying and classifying the opinions expressed in text. Earlier methods leaned on curated word lists resources like **SentiWordNet** that gave each word a fixed polarity score regardless of context. That approach has its limits, and the field has largely moved on. Transformer-based models like BERT and RoBERTa read words in relation to everything around them, which means they handle the messiness of real-world language far more reliably (Alotaibi & Dossari, 2024; Kumar et al., 2024). Aspect-Based Sentiment Analysis builds naturally on these advances.

Rather than assigning a single positive, negative, or neutral label to an entire document, ABSA identifies specific product features within the text and assesses the sentiment directed at each one independently. A smartphone review might, for example, express clear satisfaction with the camera while registering frustration about battery life in the same sentence and ABSA is designed to capture both of those opinions separately and accurately.

The value of this approach for product evaluation has been demonstrated across a number of studies. Ahmad et al. (2025) and Zhou et al. (2024) both show that ABSA significantly improves the interpretability of sentiment analysis by producing feature-level insights that document-level methods simply cannot offer. Transformer-based ABSA models have proven particularly capable when it comes to the subtler side of language sentences where sentiment is implied rather than stated outright, where irony shifts the meaning, or where opinions about several different aspects are tangled together in a single clause. That said, real challenges remain.

.Extracting relevant features accurately from informal, varied review text is difficult, and associating each sentiment expression with the correct aspect requires careful model design. These systems also depend on substantial quantities of labelled training data, and the computational demands of transformer-based models can be significant. Perhaps most importantly for this research, the literature on ABSA has concentrated predominantly on improving classification accuracy relatively little work has explored integrating ABSA into a complete, end-to-end recommendation or decision-support system, which is where this study aims to make its contribution.

3.3 Review-Driven and Explainable Recommendations

The growing volume of user-generated content online has given rise to a different breed of recommendation system one that draws on review text rather than leaning entirely on numerical ratings. The reasoning is sound: reviews carry information that star scores simply cannot. Kumar et al. (2024) show that deep learning models trained on product reviews deliver better sentiment classification and stronger predictive performance than rating-based approaches. Soni et al. (2024) make a similar case, demonstrating that web-scraped review data surfaces detailed insights into specific product features and user satisfaction that aggregate scores would never reveal.

Even so, the systems produced by this line of research share some persistent weaknesses. Most are built around maximizing prediction accuracy, with little thought given to whether users can understand or act on what they are being shown. Feature-level explanations are rarely built in. The connection between what a user says they want and the sentiments extracted from reviews tends to be loose at best, and the dominant unit of analysis remains overall sentiment rather than the granular, aspect-level breakdown that would make recommendations genuinely informative. The result is that even well-performing systems in this space often lack the structured reasoning that real-world decision-making demands.

3.4 Explainable AI in Recommendation Systems

Explainability has become one of the most actively discussed topics in modern AI, and recommendation systems sit right at the center of that conversation. The practical stakes are

clear: users who cannot understand why a system has suggested something are far less likely to trust or act on it, especially when the purchase involved is significant.

Current approaches to building explainability into recommendation systems fall broadly into three categories **rule-based explanations**, attention-based neural models, and template-based natural language generation. Zhou et al. (2024) show that pairing ABSA with explainable recommendation logic can meaningfully improve how well users understand the suggestions they receive. But most existing systems still fall short in ways that matter. They rarely integrate user-defined preferences into the explanation itself, the quantitative reasoning behind a recommendation score is seldom surfaced, and real-world deployment with genuine user evaluation remains uncommon. What explanations do exist tend to be generic rather than grounded in the actual feature data relevant to that specific user.

3.5 Literature Gap

Prior research has investigated review-based recommender systems, explainable recommendations, and Aspect-Based Sentiment Analysis (ABSA); however, these domains remain predominantly examined in isolation. Review-based recommendation studies demonstrate the significance of integrating textual reviews while also underscoring difficulties in feature extraction, review amalgamation, and transparent recommendation formulation (Hasan et al., 2025).

The literature on explainable recommendations stresses the need for user trust and openness, but many systems still depend a lot on interaction history and don't make it clear how explanations relate to clear feature-level preferences (Zhang and Chen, 2018; Ge et al., 2024). ABSA studies on product and mobile phone reviews show that they are very good at finding feature-specific sentiments, but they mostly look at how well sentiment classification works instead of how to make full decision-support systems (Sivakumar and Uyyala, 2021; Perikos and Diamantopoulos, 2024).

So, we need a smartphone recommendation system that works from start to finish, is easy to understand, and is based on features. This system should bring together ABSA, feature scoring, user-priority weighting, and user-centered evaluation into one framework.

4. Purpose of Research

4.1 Justification / Significance

This research is important because it fixes some of the biggest problems with current recommender systems and suggests a more open, requirement-driven way to recommend products.

1. Solving the Cold-Start Problem

Collaborative filtering : Most commercial recommendation systems rely on historical data to do their jobs. The system can't find patterns or make useful predictions without past ratings, click histories, and purchase records. This causes the cold-start problem, which affects both new users who haven't built up an interaction history yet and new products that haven't gotten enough reviews yet to be modeled accurately. The practical outcome is that these are exactly the kinds of situations where good recommendations would be most useful and where traditional systems fail the most.

This method completely avoids this dependency. The system can give a new user relevant, personalized suggestions from the very first session because recommendations are based on feature-level sentiment scores that are based on all user reviews, not just their own interactions. All they have to do is say which features are most important to them, like battery life, camera quality, or processing speed, and the system takes care of the rest. This makes the framework more flexible and able to grow than other options that depend on history.

2. Increasing Transparency in Recommendations

People know that modern deep learning-based recommendation systems are a problem because they are like black boxes. Most systems give up on interpretability in favor of high predictive accuracy, which is a trade-off that often happens. A user who gets a recommendation from this

kind of system has no way of knowing if it is based on something they care about or if it is just a result of patterns in the training data.

This research tackles the transparency gap by integrating ABSA with a feature-level scoring system and a natural language explanation generator. The system doesn't just show a ranked list of phones; it also explains why each one is on its top recommendations list.

A possible explanation could be:

"This smartphone is recommended because it has a long-lasting battery and a fast processor," based on real reviews that show how people really feel. That kind of clear, data-driven reasoning makes the recommendation go from being hard to understand to being clear and trustworthy.

3. Providing Feature-Based Evaluation

Overall ratings are a rough tool. A device can have a high average score across thousands of reviews, but it can also get a lot of bad reviews for one specific flaw that could make it completely useless for a certain user. For example, someone who does a lot of professional photography with their phone's camera would not be happy with a recommendation based on overall satisfaction.

This is directly addressed by the **feature-based evaluation framework** incorporated into this study. Each smartphone is evaluated separately in six important areas: camera, battery, display, performance, design, and price. The percentage of positive sentiment mentioned in the review dataset is used to determine each smartphone's performance score. As a result, each device has a structured, measurable profile that enables recommendations to be precisely matched to each user's stated priorities rather than what just so happens to be widely popular.

- Camera
- Battery
- Display
- Performance
- Design
- Price

4. Enhancing User Trust

Any AI-driven system's uptake ultimately hinges on whether or not users have enough faith in it to act upon its results. In this case, trust involves more than just accuracy; it also involves whether the logic of the system seems reasonable and approachable. The suggested system makes its reasoning visible at every stage, which is the cornerstone of interpretability and accountability, by fusing feature-level sentiment insights with transparent scoring and human-readable explanations. This is consistent with **Explainable AI's (XAI)** increasing focus on creating systems that users can comprehend and analyze.

By combining:

- Feature-level sentiment insights
- Transparent scoring mechanisms
- Human-readable explanation generation

4.2 Scope of Research

The parameters within which the suggested system is developed and assessed are established by the research's scope. The study will stay focused, attainable, and methodologically sound if the scope is clearly defined.

1. Smartphone Products Only

Within the category of consumer electronics, this study only looks at smartphones. The system is designed to examine review content specific to smartphones and assess the features that are most important in that situation. This excludes wearables, tablets, laptops, and household appliances. Limiting the domain to smartphones improves the quality of the research outputs by enabling more consistent feature extraction, more precise identification, and a more controlled evaluation environment.

2. English Reviews Only

Reviews written in English are the only ones processed and examined. This limitation is more pragmatic than philosophical: transformer models, such as BERT, perform best when tuned on a single language, and ABSA models need training data specific to that language. It would take a lot

more resources, more model training, and a lot more evaluation complexity to expand the system to support multiple languages. This study draws from a vast and varied pool of review data while maintaining a manageable implementation scope by operating in English.

3. Predefined Features

Sentiment analysis is organized around six features: camera, battery, display, performance, design, and price. These features were chosen because they are frequently mentioned in smartphone reviews and reflect the aspects that consumers value most. This version of the system lacks features like speaker quality, fingerprint sensor responsiveness, and charging speed that are more device-specific or less common. Comparing performance profiles across various devices is made simple by using a fixed, predetermined set of features, which also keeps extraction under control and consistent.

4. No Collaborative Filtering

Collaborative filtering is not used in any way by the suggested system. It does not use rating-based prediction models, similarity matrices, or user interaction histories. The user's explicit feature preferences are combined with feature-level sentiment analysis of the review text to generate all recommendations. This deliberate architectural decision removes the cold-start issue and maintains complete transparency in the system's logic.

5. No Real-Time Web Scraping

Instead of scraping real-time data while making recommendations, the system uses a pre-compiled dataset of Amazon smartphone reviews. Prior to the experimentation stage, reviews will be collected, saved locally, and utilized consistently for sentiment analysis, feature extraction, and data preprocessing. The deliberate decision to exclude real-time scraping from the system's operation was motivated by ethical and legal concerns regarding ongoing automated data collection, platform access limitations, the additional architectural complexity of live data pipelines, and the practical requirement for a reliable, consistent dataset during evaluation.

Instead of developing production-level data infrastructure, the research focuses on creating and thoroughly testing an explainable recommendation framework.

5. Objectives of the Research

5.1 Research Problem

Conventional recommendation systems mainly rely on user interaction history, collaborative filtering techniques, or overall product ratings. Although these methods may have a high degree of predictive accuracy, they frequently fall short of offering:

- Product performance insights at the feature level
- Clear and understandable explanations
- Recommendations based on requirements

Because of this, consumers get generic recommendations without knowing how particular product characteristics like battery life or camera quality affect the recommendation.

Interpretability, trust, and personalization are all diminished by this lack of transparency, particularly for users who have feature preferences.

5.2 Research Aim

The aim of this research is:

To develop an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of user reviews.

The system will analyze textual reviews, extract feature-level sentiments, compute feature performance scores, and generate transparent, requirement-driven recommendations without relying on user history.

5.3 Research Questions

This study seeks to answer the following research questions:

1. How can sentiment at the feature level be reliably extracted from reviews of smartphones?

This investigates efficient ABSA methods for sentiment classification and predefined aspect identification.

2. How can sentiment scores for features be combined to create useful performance metrics?

This looks into statistical or mathematical techniques for calculating performance percentages at the feature level.

3. How can explainable recommendations be produced using feature scores?

This investigates the conversion of sentiment-based scores into natural-language explanations and ranking systems.

4. When compared to conventional rating-based systems, does feature-based recommendation enhance interpretability?

This assesses how well feature-driven recommendations are understood and trusted by users.

5.4 Research Objectives

The following goals are established in order to fulfill the research aim:

1. To collect and prepare a structured dataset of smartphone reviews from Amazon for natural language processing and analysis
2. To identify predefined smartphone features from review text and classify the sentiment associated with each feature as positive, negative, or neutral using Aspect-Based Sentiment Analysis (ABSA).

3. To determine each smartphone's feature performance score by calculating the percentage of positive sentiment mentions for each feature.
4. To develop a recommendation engine that ranks smartphones using weighted feature scores based on user-defined feature preferences.
5. To produce AI explanation messages in plain language that support each suggestion using the actual feature scores.

6. Expected Outcomes

The following observable and quantifiable results are anticipated from this study:

1. A clean, organized dataset of Amazon smartphone reviews that is prepared for natural language processing.
2. A functional ABSA pipeline that correctly categorizes sentiment at the feature level from actual Amazon reviews.
3. A feature performance profile that displays all six of the targeted features' percentage-positive scores for every smartphone.
4. A useful recommendation engine that generates a ranked, weighted list of recommendations based on user feature priorities.
5. A user-friendly, plain-language explanation produced by AI for every suggestion.
6. When compared to a rating-based baseline, evaluation results show sentiment classification accuracy and recommendation relevance.
7. A comprehensive research report that details the project's design, execution, results, and conclusions.
8. A tidy, organized dataset of Amazon smartphone reviews is prepared for NLP analysis.

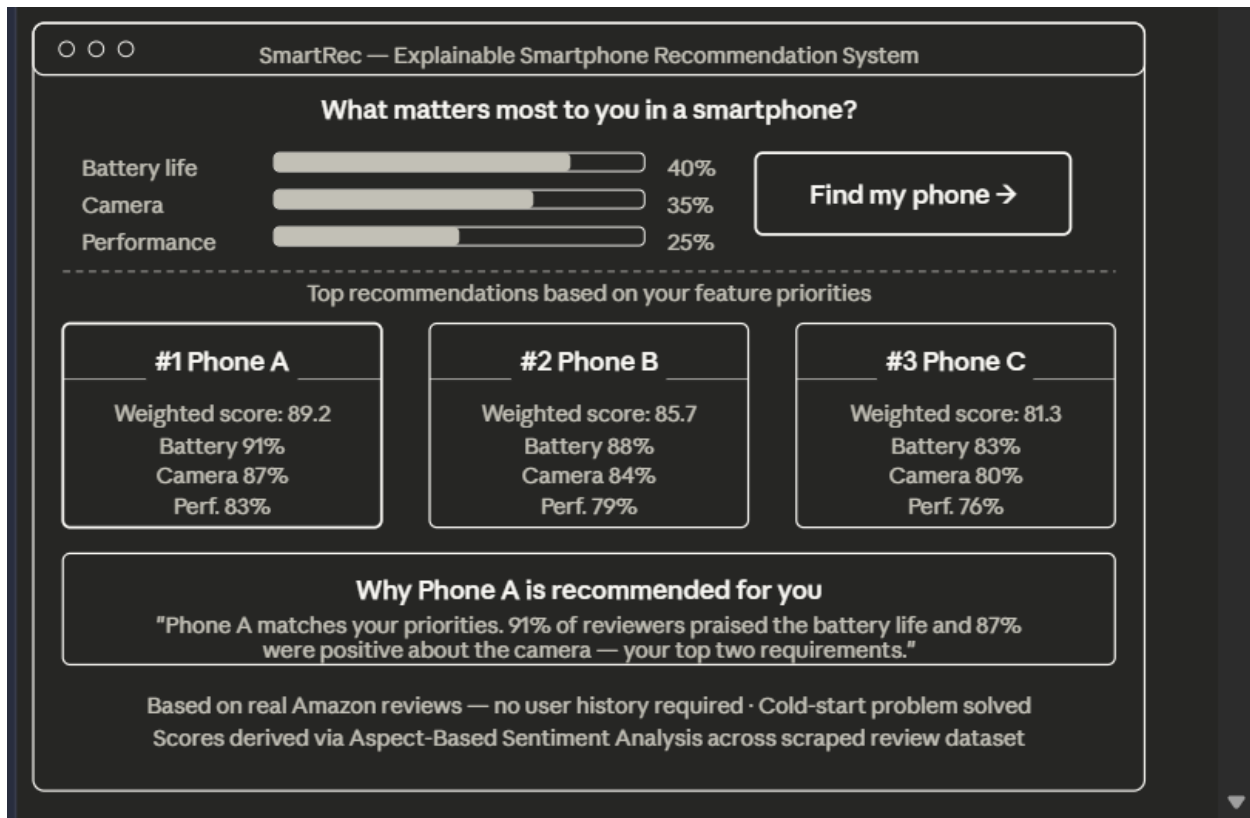


Figure 3-System Output Mock interface

7. Methodology

Since the main goal of this study is to design, implement, and assess a system that can perform feature-wise sentiment analysis (Aspect-Based Sentiment Analysis, or ABSA) on mobile phone reviews and produce explainable smartphone recommendations, it employs a Design and Development Research methodology.

Starting with data acquisition and concluding with system evaluation, the methodology adheres to a structured pipeline with seven sequential stages. To maintain a consistent and traceable

workflow throughout the research, each stage generates outputs that are used as inputs to the subsequent stage.

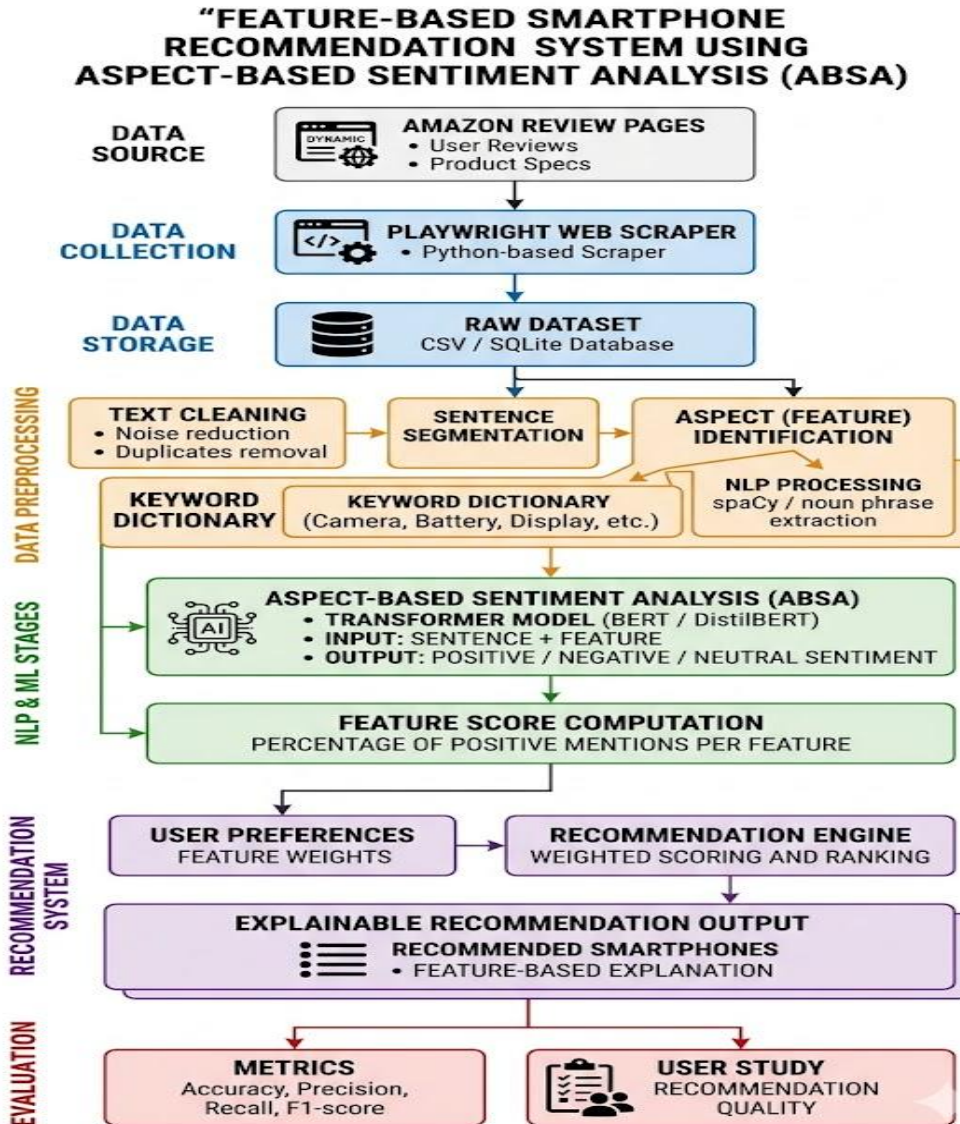


Figure 4-Research methodology flow chart

Stage 1 - Data Collection from Amazon

The first step is gathering smartphone review information from Amazon product review pages that are open to the public. Traditional static web scraping methods will be replaced with a

browser automation approach because Amazon pages are dynamically rendered and contain JavaScript-driven content.

A Python-based automation framework, Playwright, will be employed to simulate real user interactions, load dynamic content, and extract review data reliably. This method makes it possible to access dynamically loaded elements that are not available through simple HTTP requests, like review sections, pagination, and filtering options.

When available, the following characteristics will be gathered for each chosen smartphone:

- **Product name**
- **Brand**
- **Review title**
- **Review text**
- **Star rating**

Stage 2 - Data Preprocessing and Cleaning

Before analysis, the collected dataset will be preprocessed to make sure it is of good quality and consistent. There are usually problems with raw web data, such as **noise, duplicates, and inconsistencies**.

The steps for preprocessing are:

- Remove duplicate and empty entries
- Filtering out content that isn't relevant or useful
- Standardizing text by making it lowercase and taking out punctuation and special characters
- Remove too much whitespace and formatting artifacts
- Breaking up reviews into smaller parts for analysis

I will break each review down into separate sentences because feature-wise sentiment analysis works better at the sentence level. Depending on what the downstream models need, other preprocessing methods like tokenization, stopword removal, and lemmatization may be used.

Following these steps will give you a clean, organized dataset that you can use for feature extraction and sentiment classification.

Stage 3 - Feature Extraction

This stage is all about figuring out what specific smartphone features (aspects) are talked about in each sentence.

The study considers predefined aspects relevant to mobile phones, including:

- Camera
- Battery
- Display
- Performance
- Design
- Price

We will use a **hybrid method** for identifying aspects.

To begin, a keyword dictionary for each feature category will be made. For instance:

- Camera:camera, picture, image, selfie, video
- Battery:battery, charging, backup, and power
- Display: screen, resolution, and brightness

Second, tools like spaCy will be used to improve the detection of things beyond simple keyword matching by using linguistic processing techniques like noun phrase extraction and dependency parsing.

There could be more than one thing in each sentence. In these situations, each aspect mentioned will be recorded separately to make sure that all relevant opinions are included.

The result of this step is a structured dataset that connects each sentence to one or more features that have been found.

Stage 4 - Feature-Level Sentiment Classification

This stage does sentiment classification at the feature level, which is the main part of the research that analyzes data.

The system doesn't give the whole review a single sentiment; instead, it gives each feature its own sentiment.

- Positive
- Negative
- Neutral

We will use a deep learning model based on transformers, like BERT or DistilBERT, for this task. These models are chosen because they can find relationships between different parts of text, which makes them good for aspect-based sentiment analysis.

The model input will include the sentence and the identified feature, which will let the model correctly assign sentiment even when there are more than one feature in a sentence.

Example:

“The battery lasts all day, but the camera is disappointing.”

- Battery → Positive
- Camera → Negative

To improve model performance and domain relevance, a subset of the dataset will be **manually annotated** with feature-sentiment labels. This labeled data will be used for training, validation, and testing of the model.

Stage 5 - Feature Score Calculation

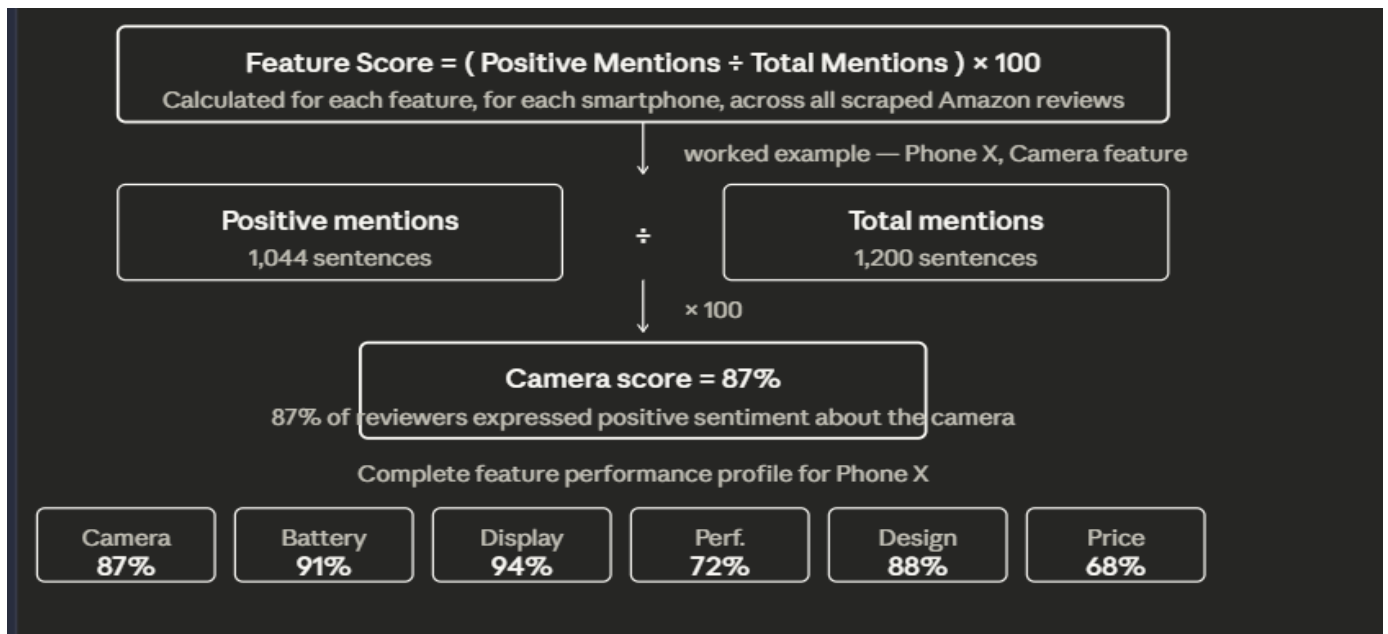


Figure 5-Feature scoring formula and worked example

Each smartphone is given a performance score based on each of the six features once all feature-level sentiments have been categorized. The formula is purposefully straightforward: to express the result as a percentage, divide the number of positive mentions of a feature by the total number of times that feature appears across all reviews of that phone, then multiply the result by 100. A phone's camera score is 86.7% if it receives 130 positive mentions out of 150 total. Every smartphone in the dataset has a comprehensive, comparable performance profile when this computation is applied to all features and devices. This is the quantitative basis upon which the recommendation engine subsequently functions.

Stage 6 - Recommendation Engine and Explanation Generation

By indicating which features are most important to them and how much weight they give each, users can interact with the system. A user who prioritizes endurance might devote 25% to performance, 35% to camera, and 40% to battery. These weights are applied to the feature scores for each device in the dataset by the engine, which then multiplies each score by the weight assigned to it before adding up the results to produce a final weighted total. The top results are then displayed as recommendations after phones are ranked from best to worst..

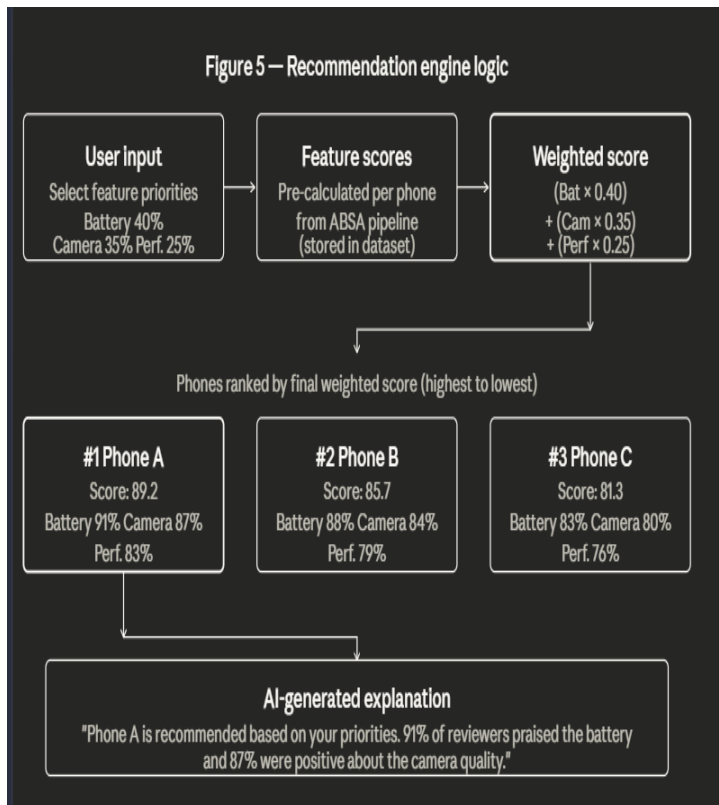


Figure 6-Recommendation engine logic

The system provides a straightforward explanation based on the actual calculated scores for each recommendation. Pre-made sentence templates with actual values are used in these explanations, such as "This smartphone is recommended based on your priorities." In 91% of reviews, users gave the battery high marks, followed by the camera (87%) and performance (83%). Any user, not just those who are familiar with data analysis, can instantly understand the recommendation thanks to the explanation.

Stage 7 - System Evaluation

The last phase evaluates the system's performance in two different dimensions. By manually labeling a sample review of sentences and comparing those labels to the model's predictions, the accuracy of sentiment classification is assessed. Accuracy, precision, recall, and F1-score are four common NLP metrics that offer a thorough understanding of classification reliability. A small user study is used to evaluate the quality of recommendations. Participants use the system, rate whether the phones that were suggested matched their stated feature priorities and determine whether the explanations were understandable and helpful. Their answers are contrasted with a baseline system that does not analyze features and only ranks phones based on their overall star rating. The comparison quantifies the benefits that the feature-driven approach offers over traditional techniques.

8. Significance of the research

For Users

Customers gain from recommendations that are clear, tailored, and based on the real experiences of customers just like them. A user can clearly see which features performed well in real reviews and understand exactly why a specific phone has appeared at the top of their results, instead of being directed toward whatever happens to be the most popular. Making confident purchasing decisions is made easier by that kind of evidence-based clarity.

For Manufacturers

This system's feature-level sentiment analysis produces structured, in-depth insight into users' opinions about aspects of individual models something that aggregate star ratings just cannot. This is directly actionable intelligence for product development teams: a comprehensive analysis, based solely on real customer feedback, of which features are generating complaints and which are receiving praise. Regardless of the system's primary recommendation function, this constitutes a significant secondary application.

For the Research Community

This work provides a useful, comprehensive illustration of what can be achieved when Explainable AI, Aspect-Based Sentiment Analysis, weighted feature scoring, and Amazon review scraping are combined into a single cohesive system. The literature currently in publication has not thoroughly examined this specific combination, and the method is both replicable and extensible. With only minor adjustments, the same architecture could be used for other consumer product categories.

9. Timeline

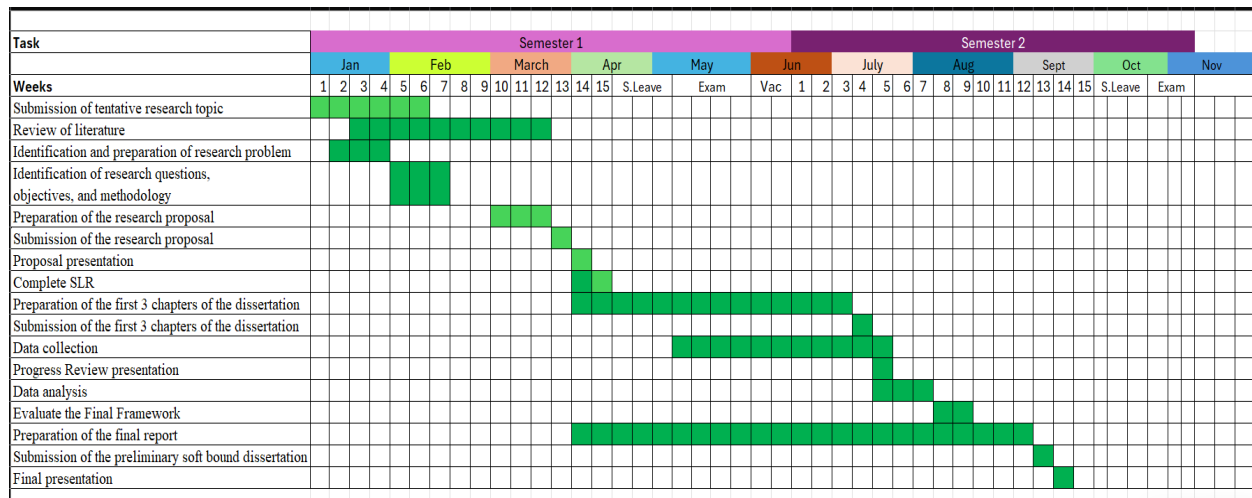


Figure 7-Timeline

10. Sustainable Development Goals (SDGs)

This research aligns with the global development agenda defined by the United Nations through the Sustainable Development Goals (SDGs). In particular, the study contributes to SDG 9 (Industry, Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production).

SDG 9: Industry, Innovation and Infrastructure

SDG 9 is all about encouraging new ideas, improving technology, and making digital infrastructure stronger. This research advances SDG 9 by developing and deploying an AI-driven recommendation system that employs Aspect-Based Sentiment Analysis (ABSA) to derive significant insights from extensive user-generated data.

The proposed system improves innovation on digital platforms by:

- Using advanced Natural Language Processing (NLP) methods like transformer-based models (BERT/DistilBERT)
- Making recommendation systems work better and smarter

- Making it possible for e-commerce businesses to make decisions based on data

This research supports the creation of smarter and more creative digital solutions by adding machine learning and explainable AI techniques to recommendation systems. This helps technology move forward and infrastructure get better.

SDG 12: Responsible Consumption and Production

SDG 12 stresses the need to encourage sustainable consumption habits and cut down on waste that isn't needed. This study supports SDG 12 by helping people choose smartphones that meet their needs and making more informed buying decisions.

Most of the time, traditional recommendation systems use overall ratings, which may not accurately show what each user likes. This research, on the other hand, presents a feature-based recommendation system that gives users suggestions based on in-depth information about product features like battery life, camera quality, and performance.

This helps people use things responsibly by:

- Helping people pick products that better fit their real needs
- Lessening dissatisfaction and unnecessary product exchanges
- Encouraging better use of technological resources

The system encourages sustainable consumption behavior by reducing impulsive or uninformed purchases by making recommendations more useful and relevant.

SDG 8 – Decent Work and Economic Growth

SDG 8 is all about encouraging long-term economic growth, boosting productivity, and helping businesses come up with new ideas. This study advances SDG 8 by creating an AI-based recommendation system that enhances the efficacy of digital commerce and facilitates improved customer decision-making on online platforms.

The suggested system helps the economy grow by:

- Making recommendation systems in e-commerce settings better and more useful

- Helping customers choose the right products based on their features
 - Helping businesses understand what their customers want by using reviews as a guide
- This research can indirectly help businesses that are focused on innovation and the digital economy by making recommendations better through sentiment analysis and explainable AI. This makes the study important for the bigger goal of making technology-enabled markets more productive and growing.

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